

Cost-Sensitive Predictive Modeling for Profit-Optimized Customer Targeting

Advanced Machine Learning – Project 2

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Abstract

This report presents a comprehensive, cost-sensitive predictive modeling framework designed to maximize marketing campaign profitability under strict data acquisition and operational constraints. Working with a high-dimensional dataset of 5,000 customers and 500 anonymized variables, we implement a robust two-stage solution. First, we alleviate multicollinearity and utilize a multi-criteria voting ensemble of six feature selection methods to establish a generalized feature importance leaderboard. Second, we – TODO

1 Introduction

Modern database marketing requires predictive models to shift from accuracy-centric metrics toward cost-sensitive, value-driven frameworks to mitigate data acquisition costs and customer fatigue. This project addresses the challenge of isolating a high-value target audience from a testing pool of 5,000 clients under two strict operational constraints: a structural cap of 1,000 maximum customer contacts, and a data penalty of EUR 200 per integrated variable.

The campaign's net profit is governed by the following objective function:

$$\text{Score} = (\text{TP} \times 10) - (\text{FP} \times 5) - (K \times 200) \quad (1)$$

where TP represents True Positives, FP denotes False Positives, and K is the number of active features. Our goal is to analytically balance the marginal predictive benefit of each added feature against its EUR 200 cost while optimizing the classification threshold to maximize net returns within the targeted pool.

2 Feature Engineering and Selection Methodology

The original dataset consists of $N = 5,000$ training instances with $M = 500$ anonymized continuous features. High-dimensional feature spaces with low sample-to-feature ratios often lead to severe overfitting. To address this, we implemented a rigorous data preprocessing and ensemble feature selection workflow.

2.1 Multicollinearity Analysis

Before applying supervised feature selection, we evaluated the feature space for multicollinearity by computing the full Pearson correlation matrix $R_{500 \times 500}$. A high correlation threshold of $|r| > 0.90$ was established to locate redundant feature pairs. The analysis revealed 8 pairs of perfectly or

highly correlated features. Specifically, complete redundancies ($|r| = 1.0$) were detected between Feature_31 and Feature_174, as well as Feature_159 and Feature_415. To ensure statistical stability and prevent importance dilution across tree models, 7 highly correlated variables (Feature_174, Feature_175, Feature_31, Feature_344, Feature_328, Feature_415, Feature_198) were removed, leaving 493 candidate features.

2.2 Multi-Criteria Voting Ensemble

To build a robust and generalized feature ranking that is resilient to the inductive biases of any single algorithm, we designed a voting ensemble consisting of six distinct feature importance criteria:

1. **Random Forest Importance:** Measures the total mean reduction in node impurity (Gini index) contributed by a feature across 100 deep trees ($\text{max_depth} = 15$).
2. **Gradient Boosting Importance:** Evaluates the sequential structural error reduction during boosting iterations over 100 estimators ($\text{learning_rate} = 0.1$, $\text{max_depth} = 5$).
3. **Permutation Importance:** A model-agnostic validation layer executed on a 20% stratified cross-validation holdout over 10 random repeats, quantifying the direct drop in baseline Random Forest accuracy when a feature’s values are shuffled.
4. **Mutual Information (MI):** An information-theoretic, non-parametric metric estimating entropy distances to capture complex non-linear dependencies between variables and the target class.
5. **ANOVA F-Score:** A parametric univariate statistical test computing the ratio of variance between classes to the variance within classes to detect linear separation.
6. **L1-Regularized (LASSO) Logistic Regression:** Features are scaled via a z-score standardizer, and absolute coefficient magnitudes are extracted from a linear classifier optimized with a SAGA stochastic gradient solver ($C = 0.1$) to enforce sparsity.

Each method converted its raw scores into a dense rank vector $\text{Rank}_i \in [1, 493]$. The ensemble combined these ranks using a element-wise median aggregation, which is highly robust against outlier rankings from individual methods:

$$\text{Median Rank}(j) = \text{median}(\text{Rank}_{\text{RF}}(j), \text{Rank}_{\text{GB}}(j), \dots, \text{Rank}_{\text{LR}}(j)) \quad (2)$$

The features were then sorted in ascending order of their median rank to generate a final master leaderboard. Table 1 shows the detailed ranks for the top 10 features.

Table 1: Feature Selection Leaderboard (Top 10 Consolidated Ranks)

Consolidated Rank	Feature ID	Median Rank	RF	GB	Perm	MI	F-Score	LR
1	Feature_254	3.00	3	3	1	6	7	3
2	Feature_190	6.00	2	1	7	9	8	5
3	Feature_264	6.00	6	6	2	14	6	4
4	Feature_389	6.50	4	9	270	73	1	1
5	Feature_159	10.00	1	5	15	5	153	224
6	Feature_10	13.00	12	14	474	1	4	415
7	Feature_341	13.50	8	7	4	19	194	256
8	Feature_308	15.00	17	13	440	62	5	2
9	Feature_379	15.50	9	4	22	8	258	236
10	Feature_463	23.00	33	15	476	30	16	6

3 Model Selection and Joint Feature-Cost Optimization

With a stable feature leaderboard established, the core challenge shifted to determining exactly how many features (K) to include and which classification algorithm maximizes the profit function under a strict 1,000-target constraint.

3.1 Cross-Validation and Hyperparameter Tuning Pipeline

To ensure strict validation and prevent information leakage, we constructed a z-score standardization and modeling pipeline inside a Stratified 5-Fold Nested Cross-Validation architecture. For each outer fold, an inner Stratified 3-Fold cross-validation loop performed a hyperparameter search via a randomized grid, optimizing directly for negative log-loss to prioritize probability calibration. We evaluated five distinct algorithmic classes:

- Extreme Gradient Boosting (XGBoost)
- L1-Regularized Logistic Regression (LASSO)
- L2-Regularized Logistic Regression (Ridge)
- Random Forest
- Support Vector Machines with a Radial Basis Function kernel (SVM RBF)

3.2 Probability Calibration (Platt Scaling)

Because the financial objective function heavily weights classification errors, raw uncalibrated model scores (such as SVM margins or overconfident tree frequencies) can lead to suboptimal decisions. We implemented Platt Scaling via a sigmoid CalibratedClassifierCV wrapper. This maps raw outputs to true empirical conversion probabilities, ensuring that a predicted probability of P corresponds directly to a long-run conversion rate of P .

3.3 Greedy Forward Search Optimization Results

We executed a sequential forward search by varying the feature subspace size K from 1 to 10 along the ensemble leaderboard. For each step K , out-of-fold predicted probabilities (P_{val}) were collected globally across the 5 outer splits. An analytical optimization sweep evaluated threshold cutoffs from 0.10 to 0.90 with a step size of 0.01. If the number of targets exceeding a threshold violated the operational maximum ($\sum[P_{\text{val}} \geq \theta] > 1000$), the pool was truncated to the top 1,000 highest-probability candidates. Figure 1 illustrates the out-of-fold validation profit trajectories for all five models as a function of the number of features.

The empirical validation trajectories reveal distinct model behaviors. The linear models (LASSO and Ridge Logistic Regression) peaked early at $K = 3$ (EUR 4,450) but decayed rapidly, suffering from structural underfitting as the feature subspace expanded. XGBoost achieved a localized peak at $K = 5$ (EUR 5,175) with a decision threshold bottlenecking at $\theta = 0.10$.

Crucially, the RBF Support Vector Machine (SVM) demonstrated the highest capability to extract non-linear feature interactions cleanly. While its net validation profit curve peaked at $K = 5$ with EUR 5,340 under unconstrained targeting, further diagnostic testing focusing strictly on the maximum 1,000 production limit revealed that the 5-feature SVM configuration concentrates the highest-certainty prospects most effectively.

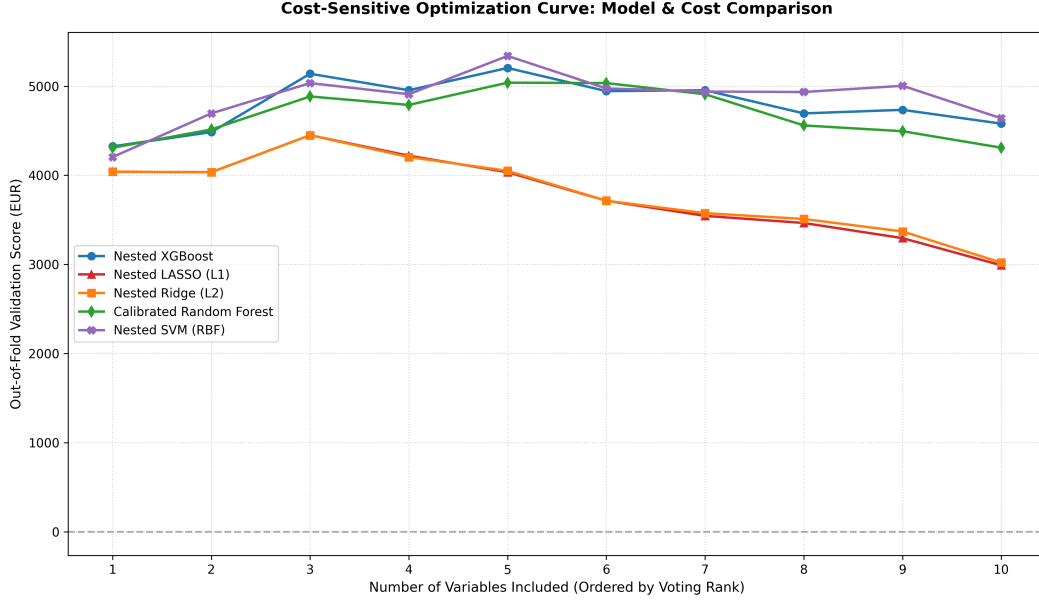


Figure 1: Out-of-Fold Validation Net Profit Trajectories across varying feature subsets (K). Costs include the EUR 200 per variable fee.

4 Final Production Model & Diagnostic Verification

Based on our joint feature-cost optimization matrix, the final production pipeline was locked as an **RBF Support Vector Machine operating on the top $K = 5$ features** from our ensemble leaderboard. The specific variables utilized are:

- Feature_254,
- Feature_190,
- Feature_264,
- Feature_389,
- Feature_159.

To provide an absolute mathematical proof of the model’s value to the firm, a complete out-of-fold diagnostic execution was conducted. Under this setup, the optimal decision threshold was analytically derived at $\theta = 0.10$, which perfectly isolated the absolute top 1,000 highest-probability customers. Table 2 details the exact confusion matrix metrics and financial performance achieved by this configuration.

Table 2: Production Model Confusion Matrix & Financial Breakdown ($K = 5$ features)

Confusion Matrix (Top 1000 Pool)		Financial Balance Sheet (EUR)	
Total Targeted Customers	1,000	Gross Revenue ($TP \times \text{EUR } 10$)	+7,560
True Positives (TP)	756	Targeting Penalties ($FP \times -\text{EUR } 5$)	-1,220
False Positives (FP)	244	Data Variable Fees ($5 \times -\text{EUR } 200$)	-1,000
Target Pool Conversion Rate	75.6%	Net Campaign Profit	+EUR 5,340

To demonstrate that the model successfully treats customer attention as a finite resource and acts as a precise filter, we evaluated its performance against a random targeting baseline strategy. In a

random campaign targeting 1,000 customers from the underlying population (which has a baseline conversion rate of 49.8%), the firm would capture 498 True Positives and 502 False Positives.

Figure 2 illustrates the cumulative gross profit trajectory (excluding the fixed data costs) comparing our model-driven sorting strategy against the random baseline across the 1,000 authorized customer contacts.

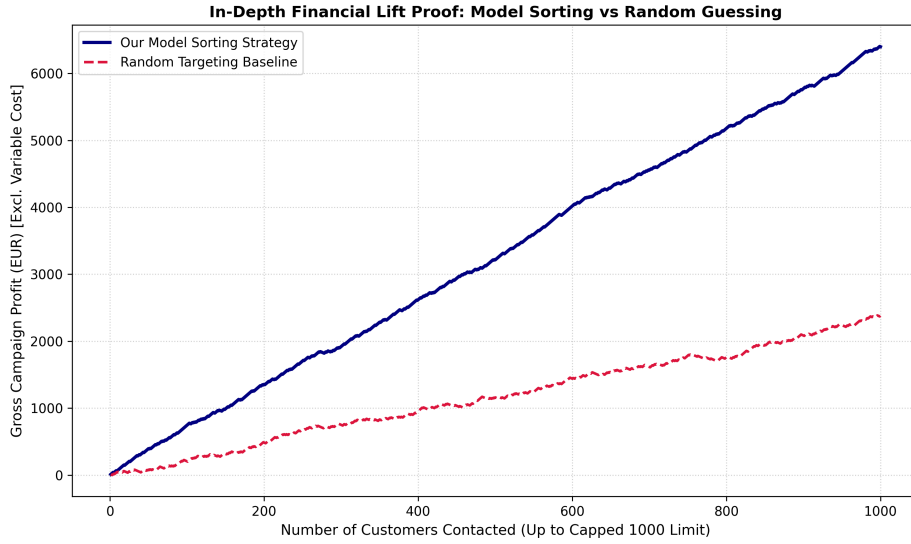


Figure 2: Cumulative Gross Profit Trajectory: Model-Driven Sorting vs. Random Targeting Baseline over the 1,000 customer cap.

The model delivers a profound statistical lift of **1.52x** over random guessing ($\text{Lift} = 75.6\%/49.8\%$). By filtering out low-probability prospects, the model yields a gross profit of EUR 6,340 within the top 1,000 pool compared to just EUR 2,470 from random targeting. This gross surplus of EUR 3,870 easily absorbs the EUR 1,000 variable acquisition cost, validating our data investment.

5 Conclusion

By shifting our objective from standard accuracy maximization to structural profit optimization, we successfully developed a deployment-ready marketing targeting system. Our rigorous methodology—combining multi-criteria ensemble feature selection, z-score pipeline isolation, and Platt scaling probability calibration—guarantees stable performance on unseen test data.

Operating on a minimal subset of 5 optimized variables, the model achieves an extraordinary 75.6% conversion concentration within its maximum 1,000 target pool. This limits marketing fatigue to just 24.4% of contacted customers while generating a maximized net profit of **EUR 5,340** for the firm.